

# A - Secret Numbers

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

## Problem Statement

You are given a string  $S$  consisting of lowercase English letters and digits.

Extract only the digit characters from  $S$  and output the string formed by arranging them in their original order.

## Constraints

- $S$  is a string of length between 1 and 50, inclusive, consisting of lowercase English letters and digits.

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## Input

The input is given from Standard Input in the following format:

$S$

## Output

Output the answer.

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## Sample Input 1

abc462

## Sample Output 1

462

The digits contained in abc462 are 4, 6, 2. Concatenating them in their original order gives 462. Thus, output 462.

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## Sample Input 2

codequeen

## Sample Output 2

codequeen contains no digits, so output the empty string.

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## Sample Input 3

31415

## Sample Output 3

31415

## Sample Input 4

10plus2is12

## Sample Output 4

10212

# B - Gift

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

## Problem Statement

$N$  people, numbered 1 through  $N$ , exchanged gifts with each other.

Person  $i$  sent gifts to  $K_i$  people: persons  $A_{i,1}, A_{i,2}, \dots, A_{i,K_i}$ .

For each  $i = 1, 2, \dots, N$ , find all people who sent a gift to person  $i$ .

## Constraints

- $2 \leq N \leq 100$
  - $1 \leq K_i \leq N - 1$
  - $1 \leq A_{i,1} < A_{i,2} < \dots < A_{i,K_i} \leq N$
  - $A_{i,j} \neq i$
  - All input values are integers.
- 

## Input

The input is given from Standard Input in the following format:

```
N
K_1 A_{1,1} A_{1,2} ... A_{1,K_1}
K_2 A_{2,1} A_{2,2} ... A_{2,K_2}
⋮
K_N A_{N,1} A_{N,2} ... A_{N,K_N}
```

## Output

Output  $N$  lines.

For the  $i$ -th line, let  $B_1, B_2, \dots, B_X$  be the numbers of the people who sent a gift to person  $i$ , listed in ascending order (where  $X$  is the count of people who sent a gift to person  $i$ ), and output in the following format:

```
X B_1 B_2 ... B_X
```

---

## Sample Input 1

```
4
1 2
1 3
1 2
3 1 2 3
```

## Sample Output 1

```
1 4
3 1 3 4
2 2 4
0
```

Person 1 sent a gift to person 2, person 2 sent a gift to person 3, person 3 sent a gift to person 2, and person 4 sent gifts to persons 1, 2, 3.

Person 1 received a gift from person 4. Thus, output 1 4 on the first line.

Person 2 received gifts from persons 1, 3, 4. Thus, output 3 1 3 4 on the second line.

Person 3 received gifts from persons 2, 4. Thus, output 2 2 4 on the third line.

Person 4 did not receive any gifts. Thus, output 0 on the fourth line.

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## Sample Input 2

```
4
3 2 3 4
2 1 4
2 1 2
2 2 3
```

## Sample Output 2

```
2 2 3
3 1 3 4
2 1 4
2 1 2
```

---

## Sample Input 3

```
7
1 3
4 3 4 6 7
1 7
3 2 6 7
2 3 7
1 4
1 5
```

## Sample Output 3

```
0
1 4
3 1 2 5
2 2 6
1 7
2 2 4
4 2 3 4 5
```

# C - Not Covered Points

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

## Problem Statement

There are  $N$  points numbered 1 through  $N$  on a two-dimensional plane. Point  $i$  ( $1 \leq i \leq N$ ) has coordinates  $(X_i, Y_i)$ . Here, it is guaranteed that  $X$  and  $Y$  are each permutations of  $(1, 2, \dots, N)$ .

Find the number of values of  $i$  such that the interior (not including the boundary) of the rectangle with sides parallel to the  $x$ -axis and  $y$ -axis, with lower-left vertex  $(0, 0)$  and upper-right vertex  $(X_i, Y_i)$ , contains none of the  $N$  points numbered 1 through  $N$ .

## Constraints

- $1 \leq N \leq 3 \times 10^5$
- $1 \leq X_i, Y_i \leq N$
- $X$  and  $Y$  are each permutations of  $(1, 2, \dots, N)$ .
- All input values are integers.

---

## Input

The input is given from Standard Input in the following format:

```
N
X1 Y1
X2 Y2
⋮
XN YN
```

## Output

Output the answer.

---

## Sample Input 1

```
3
2 1
1 3
3 2
```

## Sample Output 1

```
2
```

The two values  $i = 1, 2$  satisfy the condition.

For  $i = 3$ , the interior (not including the boundary) of the rectangle with sides parallel to the  $x$ -axis, with lower-left vertex  $(0, 0)$  and upper-right vertex  $(3, 2)$ , contains point 1.

---

## Sample Input 2

```
5
1 1
4 2
2 3
5 5
3 4
```

## Sample Output 2

```
1
```

## Sample Input 3

```
7
3 4
6 1
5 5
2 7
7 2
1 3
4 6
```

## Sample Output 3

```
2
```



# D - Accomplice

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

## Problem Statement

A murder occurred at a certain mansion. There are  $N$  suspects, called person 1, person 2, . . . , person  $N$ .

Person  $i$  entered the mansion at time  $S_i$ , left the mansion at time  $T_i$ , and did not enter or leave at any other times.

The following facts are known about the crime:

- There are exactly two culprits.
- The crime started at some integer time  $x$ , took  $D$  units of time, and was completed at time  $x + D$ .
- Both culprits were in the mansion at all times from the start to the completion of the crime. (They may have entered the mansion exactly when the crime started, or left exactly when it was completed.)

Assuming that both culprits are among the  $N$  suspects, how many combinations of the two culprits and the crime start time are possible? Here, the order of the two culprits does not matter.

## Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq S_i \leq T_i \leq 10^6$
- $1 \leq D \leq 10^6$
- All input values are integers.

---

## Input

The input is given from Standard Input in the following format:

```
 $N$   $D$   
 $S_1$   $T_1$   
 $S_2$   $T_2$   
 $\vdots$   
 $S_N$   $T_N$ 
```

# Output

Output the number of possible combinations of the two culprits and the crime start time.

---

## Sample Input 1

```
3 2
9 17
10 12
13 20
```

## Sample Output 1

```
4
```

In this sample, there are three suspects, and the crime took two units of time.

- If persons 1 and 2 are the culprits, they were both in the mansion from time 10 to time 12. If the crime start time is time 10, the crime can be committed in two units of time.
- If persons 1 and 3 are the culprits, they were both in the mansion from time 13 to time 17. If the crime start time is 13, 14, or 15, the crime can be committed in two units of time.
- If persons 2 and 3 are the culprits, they were never in the mansion at the same time, so this pair cannot have committed the crime.

Thus, the possible combinations of the two culprits and the crime start time are (persons 1, 2, time 10), (persons 1, 3, time 13), (persons 1, 3, time 14), (persons 1, 3, time 15), giving four combinations.

---

## Sample Input 2

```
3 5
9 17
10 12
13 20
```

## Sample Output 2

```
0
```

The suspects are the same as in Sample Input 1, but this time the crime took five units of time.

No matter which two persons are the culprits, the time they were both in the mansion is shorter than the time required for the crime, so the crime is impossible.

Thus, the number of possible combinations of the two culprits and the crime start time is zero. (In other words, our assumption is wrong, and the real culprit has escaped us.)

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## Sample Input 3

```
4 1
1 1000000
1 1000000
1 1000000
1 1000000
```

## Sample Output 3

```
5999994
```

Beware of overflow for larger cases.

# E - Alternating Costs

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

## Problem Statement

You are given integers  $A, B, X, Y$ .

A piece is placed on a two-dimensional plane. Initially, the piece is at coordinate  $(0, 0)$ .

You can perform the following operation zero or more times:

- Let the current coordinate of the piece be  $(x, y)$ . Move the piece to one of the coordinates  $(x - 1, y)$ ,  $(x + 1, y)$ ,  $(x, y - 1)$ ,  $(x, y + 1)$ .

The cost of the  $k$ -th operation ( $k \geq 1$ ) depends on the parity of  $k$ , as follows:

- If  $k$  is odd: Let the current coordinate of the piece be  $(x, y)$ . The cost of moving to  $(x - 1, y)$  or  $(x + 1, y)$  is  $A$ , and the cost of moving to  $(x, y - 1)$  or  $(x, y + 1)$  is  $B$ .
- If  $k$  is even: Let the current coordinate of the piece be  $(x, y)$ . The cost of moving to  $(x - 1, y)$  or  $(x + 1, y)$  is  $B$ , and the cost of moving to  $(x, y - 1)$  or  $(x, y + 1)$  is  $A$ .

Find the minimum total cost required to move the piece to coordinate  $(X, Y)$ .

$T$  test cases are given; solve each one.

## Constraints

- $1 \leq T \leq 2 \times 10^5$
  - $1 \leq A, B \leq 10^9$
  - $-10^9 \leq X, Y \leq 10^9$
  - All input values are integers.
-

## Input

The input is given from Standard Input in the following format:

```
 $T$   
case1  
case2  
⋮  
case $T$ 
```

The  $i$ -th ( $1 \leq i \leq T$ ) test case case <sub>$i$</sub>  is given in the following format:

```
 $A$   $B$   $X$   $Y$ 
```

## Output

Output the answers for the test cases in order, each on a separate line.

### Sample Input 1

```
5  
1 2 -1 2  
8 5 0 0  
7 13 9 4  
1 1 0 100  
31 9 -74 -60
```

### Sample Output 1

```
4  
0  
103  
100  
1332
```

Consider the first test case.

The following moves cost 4 in total:

- Move the piece from  $(0, 0)$  to  $(-1, 0)$ . This costs 1.
- Move the piece from  $(-1, 0)$  to  $(-1, 1)$ . This costs 1.
- Move the piece from  $(-1, 1)$  to  $(-1, 2)$ . This costs 2.

It is impossible to move the piece to  $(-1, 2)$  with a total cost less than 4, so output 4 on the first line.



# F - More ABC

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 500 points

## Problem Statement

You are given a string  $S$  consisting of uppercase English letters.

Takahashi's goal is to **increase** the number of occurrences of ABC as a substring of  $S$  by exactly  $K$ , by replacing some characters of  $S$ .

Determine whether this is possible, and if so, find the minimum number of characters Takahashi needs to replace to achieve his goal.

$T$  test cases are given; solve each one.

► What does "replacing" mean?

► What is a substring?

## Constraints

- $1 \leq T \leq 10^5$
  - $S$  is a string of length between 3 and  $3 \times 10^5$ , inclusive, consisting of uppercase English letters.
  - $1 \leq K \leq 10$
  - In each input, the total length of  $S$  over all test cases is at most  $3 \times 10^5$ .
  - $T$  and  $K$  are integers.
-

# Input

The input is given from Standard Input in the following format:

```
 $T$   
case1  
case2  
⋮  
case $T$ 
```

case <sub>$i$</sub>  represents the  $i$ -th test case.

Each test case is given in the following format:

```
 $S$   
 $K$ 
```

# Output

Output  $T$  lines.

The  $i$ -th line ( $1 \leq i \leq T$ ) should contain the answer for the  $i$ -th test case.

For each test case, if it is impossible for Takahashi to achieve his goal, output  $-1$ ; otherwise, output the minimum number of characters that need to be replaced to achieve his goal.

---

## Sample Input 1

```
5  
ATCABC  
1  
ABABC  
1  
XABCYZ  
1  
ZZZZZ  
1  
ABCBABCAEFCABAABC  
2
```

## Sample Output 1

```
1
-1
6
3
3
```

In the first test case, the given string  $S = \text{ATCABC}$  contains ABC as a substring once.

Replacing the second character of  $S$  with B gives  $S = \text{ABCABC}$ , which contains ABC as a substring twice.

Thus, replacing one character increases the number of occurrences of ABC as a substring by 1, achieving the goal. Therefore, output 1 on the first line.

In the second test case, the given string  $S = \text{ABABC}$  contains ABC as a substring once; no matter how the characters of  $S$  are replaced, it is impossible to make  $S$  contain ABC as a substring twice.

Therefore, output  $-1$  on the second line.

In the third test case, the given string  $S = \text{XABCYZ}$  contains ABC as a substring once.

To make  $S$  contain ABC as a substring twice, all characters of  $S$  must be replaced, giving  $S = \text{ABCABC}$ .

Therefore, output 6 on the third line.

# G - Completely Wrong

Time Limit: 3 sec / Memory Limit: 1024 MiB

Score : 575 points

## Problem Statement

There is a box containing  $N$  balls numbered 1 through  $N$ . Each ball is painted with a color represented as an integer, and ball  $i$  is painted with color  $C_i$ .

Takahashi performed  $N$  operations. The  $k$ -th operation ( $1 \leq k \leq N$ ) is as follows:

- Draw one ball uniformly at random from the box and discard it. If the drawn ball has color  $G_k$ , gain 1 point.

Find the probability, modulo 998244353, that the total score over  $N$  operations is 0 points.

► Definition of probability modulo 998244353

## Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq C_i, G_k \leq N$
- All input values are integers.

## Input

The input is given from Standard Input in the following format:

```
N
C_1 C_2 ... C_N
G_1 G_2 ... G_N
```

## Output

Output the sought probability modulo 998244353.

## Sample Input 1

```
3
1 2 3
1 1 2
```

## Sample Output 1

```
332748118
```

For example, the operations may proceed as follows:

- First operation: A ball with color 2 is drawn from the box.
- Second operation: A ball with color 1 is drawn from the box. Gain 1 point.
- Third operation: A ball with color 3 is drawn from the box.

In this case, the total score is 1 point.

The probability that Takahashi's total score is 0 is  $\frac{1}{3}$ .

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## Sample Input 2

```
3
1 1 3
3 3 3
```

## Sample Output 2

```
0
```

Takahashi's total score is always 1 point.

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## Sample Input 3

```
6
1 2 3 3 4 5
3 5 3 5 5 4
```

## Sample Output 3

```
316110712
```